Arista Access Node BAS-ACC-10 Basavanagudi POP Hardware/Environmental Fault ALARM_5365 Critical Severity SOP

Node: BAS-ACC-10 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_5365

Description

The failure domain of Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) on the Arista Access Node BAS-ACC-10 is attributed to the vulnerability of the Arista 7250X series line cards to memory parity errors due to the high-speed operation of the 100GbE and 400GbE interfaces. Additionally, the loss of power supply redundancy can stem from the failure of the redundant power supplies or a misconfiguration of the redundant power supply settings. This failure mode can also be exacerbated by environmental factors such as high temperatures, humidity, or power fluctuations. In this SOP, we will guide you through the steps to diagnose and resolve the Hardware/Environmental Fault on the Arista Access Node BAS-ACC-10 at the Basavanagudi POP. The diagnosis and resolution steps will focus on identifying and resolving the root cause of the failure, which may involve inspecting the line cards, power supplies, and environmental conditions.

Pre-requisite Checks

- Prerequisite: Verify the physical connection to the Basavanagudi POP and ensure that the remote hands are dispatched to the site., Status: Completed

- Prerequisite: Check the redundant link status to ensure that it is up and functioning properly., Status: Completed

- Prerequisite: Verify the power supply redundancy settings and ensure that they are correctly configured., Status: Completed

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to high-speed interface operation	High
Line card crashes due to software or hardware failure	Medium
Loss of power supply redundancy due to faulty power supplies or misconfiguration	Medium
Environmental factors such as high temperatures, humidity, or power fluctuations	Low

Diagnostic Commands & Expected Outputs

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Command: show system internal line-card info, Expected Output: Line card information including memory parity errors and crash counts, Description: This command will display information about the line
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cards, including any memory parity errors or crash counts. This will help identify if the issue is related to the line cards.

Command: show system internal power-info, Expected Output: Power supply information including redundancy status and power consumption, Description: This command will display information about the power supplies, including the redundancy status and power consumption. This will help identify if the issue is related to the power supplies.

Command: show environment temperature, Expected Output: Temperature information including ambient temperature and temperature thresholds, Description: This command will display information about the ambient temperature, including any temperature thresholds that may be exceeded. This will help identify if the issue is related to environmental factors.
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Resolution Steps

Step 1: Step: Inspect the line cards for any signs of physical damage or wear, Command: show system internal line-card info, Description: Inspect the line cards for any signs of physical damage or wear. If any issues are found, replace the line card with a new one.

Step 2: Step: Verify the power supply redundancy settings and correct any misconfiguration, Command: config t set system power-supply redundancy 1 exit, Description: Verify the power supply redundancy settings and correct any misconfiguration. Ensure that the power supply redundancy is properly configured to prevent loss of redundancy.

Step 3: Step: Replace the faulty power supplies with new ones, Command: show system internal power-info replace power-supply 1 with new show system internal power-info, Description: Replace the faulty power supplies with new ones. Ensure that the new power supplies are properly installed and configured.

Post-Check Verification

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Command: show system internal line-card info, Expected Output: Line card information including memory parity errors and crash counts, Description: Verify that the line card information shows no memory parity errors or crash counts.

Command: show system internal power-info, Expected Output: Power supply information including redundancy status and power consumption, Description: Verify that the power supply information shows proper redundancy status and power consumption.

Command: show environment temperature, Expected Output: Temperature information including ambient temperature and temperature thresholds, Description: Verify that the ambient temperature is within the safe threshold.
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Roll-back Procedures

- Step: Restore the previous configuration, Command: show run rollback 1, Description: Restore the previous configuration by rolling back the configuration to the previous commit.

- Step: Replace the faulty power supplies with the original ones, Command: show system internal power-info replace power-supply 1 with original show system internal power-info, Description: Replace the faulty power supplies with the original ones. Ensure that the original power supplies are properly installed and configured.